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1. Introduction

Agriculture remains the backbone of Sri Lanka's economy, with paddy cultivation occupying a central role in ensuring national food security. However, the sector faces mounting challenges due to limited water resources, erratic rainfall patterns, and the high cost of irrigation management. Traditional irrigation practices for paddy fields largely depend on farmers' experience, manual observations, and fixed schedules. These approaches often lead to either over-irrigation, which wastes water and depletes reservoirs, or under-irrigation, which stresses crops and reduces yield. The consequences of these inefficiencies are particularly severe in developing countries, where the agricultural sector already contends with climate change, rising input costs, and increasing competition for freshwater resources from urban and industrial sectors.

Recent advances in precision agriculture, Internet of Things (IoT) devices, and machine learning offer a promising pathway to address these challenges. Globally, research has demonstrated the potential of soil moisture sensors, weather forecasting, and automated irrigation systems to improve water-use efficiency. However, a critical gap exists in integrating these components into a single, affordable, and field-ready solution specifically designed for paddy cultivation. Many existing studies focus either on soil moisture prediction models or on automated water gate mechanisms in isolation. Furthermore, most solutions rely on expensive hardware or do not incorporate forward-looking weather data, such as rainfall forecasts and temperature projections, which are essential for proactive irrigation planning.

This proposed project aims to fill that gap by developing a **Smart Water Gate System for Paddy Cultivation** a low-cost, AI-driven irrigation management solution that combines real-time environmental sensing, predictive modeling, and automated actuation. The system will use sensors to monitor soil moisture and water tank levels, while free weather APIs will provide temperature and rainfall forecasts. A machine learning model will then process these inputs to predict the optimal soil moisture levels required at a given time. Based on these predictions, a motorized gate mechanism will automatically regulate water flow into the paddy field.

By integrating real-time data with predictive analytics, the proposed system moves beyond reactive irrigation toward a proactive and intelligent water management strategy. This innovation addresses an urgent need for resource-efficient farming practices that can enhance productivity, reduce environmental impact, and improve farmers' resilience to climate variability. Ultimately, the project contributes to closing a significant technological gap in the field of smart irrigation systems for paddy cultivation, offering a scalable and practical framework that can be adapted across similar agricultural contexts in Sri Lanka and beyond.

2. Research Problem and Objectives

Research Problem

Efficient water management in paddy cultivation remains a major challenge in Sri Lanka and other rice-producing countries. Farmers still rely on traditional irrigation methods, which are often based on fixed schedules or subjective judgments rather than real-time data. While soil moisture sensors, weather forecasts, and automated water gates have each been studied individually, there is no system integrate all three components into a unified, low-cost solution suitable for smallholder farmers. Existing research on soil moisture prediction models focuses mainly on single data sources (e.g., soil sensors) and does not combine future weather predictions to anticipate irrigation needs. Similarly, most automated gate control systems act reactively opening or closing gates based on instantaneous thresholds rather than proactively using predictive analytics. This results in water wastage, inconsistent soil moisture, and reduced crop yields.

This project addresses this clear gap by developing a Smart Water Gate System that integrates real-time sensor data (soil moisture and water tank levels) with free weather API forecasts (rainfall and temperature) to predict and deliver the required irrigation automatically through a motorized gate mechanism.

Research Questions

1. How can real-time soil moisture, water tank level, and weather forecast data be combined to predict irrigation requirements for paddy cultivation?
2. Which machine learning approach best predicts the optimal soil moisture levels and irrigation needs in real time?
3. How can the predictions be reliably translated into automated gate control actions in a low-cost hardware setup?
4. What measurable improvements in water efficiency and crop health can be achieved compared to conventional irrigation practices?
5. What factors affect soil moisture in paddy fields, and how should these be considered in prediction models?
6. What types of hardware and software have been used in past projects on soil moisture prediction and smart irrigation systems, and what limitations do they present?

Objectives

- **O1:** Design and implement a low-cost, sensor-driven smart irrigation system for paddy fields using Raspberry Pi 4 Model B.
- **O2:** Integrate soil moisture, water tank level, and weather forecast data (rainfall and temperature) into a single predictive framework.
- **O3:** Develop and train a machine learning model to estimate real-time water requirements for paddy cultivation.

- **O4:** Link the predictive model output to a motorized gate mechanism for automatic irrigation control.
- **O5:** Evaluate system performance in terms of water savings, prediction accuracy, and impact on crop yield compared to traditional methods.

3. Literature Review

Area	Given Paper	My Proposed Smart Water Gate System	Technological Gap
Soil Moisture Prediction using ANN (Arif et al., 2013)	Predicts soil moisture using ANN with limited weather inputs (temperature, precipitation, evapotranspiration). Achieved $R^2 = 0.80$ (training), $R^2 = 0.73$ (validation). No real-time sensing or irrigation automation.	Predicts **Current Water Requirement (CWR)** using real-time soil-moisture, tank-level, and rain-forecast data. Automates irrigation via a **motorized gate system** executing model-driven decisions.	Lacks **real-time sensing**, **forecast-based prediction**, and **automated irrigation control**. Only estimates soil moisture; does not convert prediction into field-level irrigation action.
Satellite-Based Soil Parameter Mapping (Ghazali et al., 2019)	Uses Landsat-8 imagery and regression models to estimate soil salinity, soil moisture, and soil pH across paddy fields. Provides monitoring insights but no real-time or control mechanisms.	Integrates **ground sensors** for soil moisture, tank-water level, and rainfall forecast to predict CWR. Includes **automated gate actuation** for real-time irrigation control.	Relies on **coarse, non-real-time satellite data**; no irrigation feedback loop. Models are static and not adaptive to dynamic field or rainfall changes.
IoT-Based Smart Irrigation using ML (Raja et al., 2022)	Combines IoT sensors (moisture, humidity, temperature) with ML models (RF, SVM, NB, XGBoost) to decide irrigation via solenoid-valve control. Achieves 95.6 % accuracy with RF.	Uses real-time sensors plus **rain prediction** and **tank-level monitoring** to predict quantitative CWR. Employs **motorized gate** for large-scale paddy-field irrigation.	Existing system focuses on **classification-based irrigation timing**, not quantitative CWR; lacks paddy-specific water-logging logic and mechanical gate integration.
Optical Remote-Sensing Soil-	Estimates Soil Moisture Index (SMI) from Landsat	Uses **real-time sensor network** (soil	**Remote-sensing approach is**

Moisture Estimation (Sholihah et al., 2022)	8 NDVI and LST to map paddy-field moisture. Provides periodic spatial insights but no control or forecasting.	moisture + tank level + rain forecast) for continuous CWR prediction and automated gate actuation.	**passive**—limited by revisit intervals and spatial resolution. No dynamic forecasting, actuation, or local-level control loop.
Low-Cost Edge ML for Soil Moisture Prediction (Ramesh Kumar et al., 2022)	Performs soil-moisture prediction using ML (RF, ANN) on edge devices (Raspberry Pi) for real-time inference. Focuses on affordability and reducing cloud dependency.	Extends the edge-based idea with **CWR prediction** using real-time data and **motorized gate actuation** for autonomous irrigation. Integrates **rain forecast** and **tank-level awareness** for paddy-specific optimization.	The paper emphasizes **prediction only**; lacks irrigation feedback, paddy adaptation, and environmental awareness. No mechanical control or automation implemented.
Optimized ML for Soil Moisture Prediction (Nirala et al., 2020)	Uses optimized ML (Firefly + ANN, SVM, RF) for precise soil-moisture prediction. Achieved 98% accuracy. Input data from weather stations (temperature, humidity, etc.). Focused purely on improving prediction accuracy.	Extends beyond prediction to **real-time irrigation control** through a smart gate system. Incorporates **rain prediction**, **tank-level sensing**, and **temporal forecasting** to predict actual CWR.	Focuses solely on algorithm optimization without real-world automation or environmental integration. Lacks **forecasting, actuation, and crop-specific adaptation** for paddy irrigation.
Hybrid ML for Short-Term Soil Moisture Forecasting (Shamshirband et al., 2018)	Combines M5 model tree with Firefly optimization to forecast short-term soil moisture (24–48 hrs) using rainfall, humidity, evaporation, and temperature. Delivers high predictive accuracy but only analytical in nature.	Incorporates **dynamic learning models** for **CWR forecasting** using sensor fusion (soil moisture, rain forecast, tank level) and **actuates a motorized water gate** for autonomous irrigation control.	The paper is **forecast-only** with no irrigation automation or feedback integration. It lacks **real-time adaptability, actuation, and environmental resource management** for specific crops like paddy.

Deep Learning for Soil Sensor Development (Zhang et al., 2022)	Focuses on improving soil-sensor calibration using deep learning (CNNs, autoencoders, LSTMs). Targets soil property detection via spectral or imaging data. Systems are diagnostic and lab-based.	Utilizes **multi-sensor fusion** (soil, tank, rainfall) for predictive irrigation control. Connects AI analysis directly to **smart gate actuation** for paddy irrigation.	The paper improves sensing accuracy but remains **non-predictive and non-autonomous**. No **forecasting, actuation, or real-time control** mechanism for irrigation.
Root-Zone Moisture Estimation using AutoML (Babaeian et al., 2021)	Estimates root-zone soil moisture using UAV imagery, AutoML (NN, GBM, GLM), and soil physical properties. Achieved $R^2 > 0.90$ using NDVI and hydraulic soil data. Focused on mapping and analysis.	Predicts **CWR** using real-time field sensors (soil, tank, forecast) and **automatically actuates** irrigation through a motorized water gate.	Provides accurate root-zone estimation but lacks **real-time prediction**, **automation**, and **paddy-specific adaptation**. No decision layer or irrigation control mechanism.
Multi-Layer Soil Moisture Estimation using Ensemble Learning (Srivastava et al., 2021)	Estimates root-zone soil moisture at multiple depths (0–100 cm) using Random Forest, SVR, and ANN on satellite data (SMAP, ESA-CCI, NDVI, rainfall). Generates high-resolution maps ($R^2 > 0.8$).	Uses **surface-level sensors** and **rain + tank-level data** to predict **CWR** and control a motorized gate for paddy irrigation.	Focuses on multi-depth moisture mapping, not irrigation. Relies on satellite data—**not real-time**, **no control integration**, and **not deployable** for small-scale field irrigation.
Automatic Gate Controlling System for Sustainable Agriculture (Rathnayaka et al., 2024)	Proposes a **PID-based smart irrigation system** with soil and water sensors, solar-powered ESP32 modules, and motorized sluice gates for paddy irrigation in Sri Lanka. Enables feedback control and remote monitoring through dashboards.	Enhances this design with **AI-driven prediction** of irrigation need (CWR), **rain forecast integration**, and **crop-stage adaptability** to improve efficiency and intelligence.	The reviewed system is **automated but reactive**, lacking **predictive intelligence**, **rain-awareness**, and **crop-specific optimization** — limiting proactive irrigation scheduling.
Automated Water-Gate Controlling	Implements an IoT and PID-based water management system using	Advances beyond PID logic by adding **AI-driven CWR**	While operationally complete, it lacks **intelligent**

System for Paddy Fields (Sanjula et al., 2020)	sensor nodes, reservoir and paddy gates, and ESP-NOW wireless communication. Maintains 8 cm water level and predicts short-term rain via SVM. Provides full automation from sensing to actuation.	**prediction**, **dynamic weather forecasting**, and **crop-stage adaptive control** for optimized irrigation scheduling.	**prediction**, **dynamic adaptability**, and **forecast generalization**. System is rule-based and reactive, not proactive or learning-enabled.
Automatic Watergate System Using Fuzzy Logic (Rama et al., 2021)	Uses **fuzzy logic control** on ESP32 with water level and soil moisture sensors to determine gate opening levels (0%, 50%, 100%). Employs Mamdani inference and radio-based wireless communication. Demonstrated field reliability and flexibility.	Employs **AI-driven predictive modeling** integrating real-time data, rainfall forecasts, and crop-specific irrigation logic to determine dynamic water requirement.	Fuzzy logic system is **reactive, not predictive**, lacking environmental forecasting, adaptive learning, and dynamic scheduling. Cannot optimize irrigation quantity or account for upcoming rainfall.
Gate Guard: Smart Water Gate Automation (Niharika et al., 2024)	IoT-based water gate control using Arduino, water level sensor, servo motor, GSM alerts, and cloud dashboard via ThingSpeak. Sends SMS notifications when water level exceeds threshold. Provides real-time monitoring and automatic gate actuation.	Integrates **AI-based irrigation prediction** using multi-sensor inputs (soil, water, forecast). Automates gate movement with **proactive scheduling** instead of static threshold rules.	Limited to **threshold-based decisions**, **no forecasting or learning**, and **single-sensor dependency**. Lacks adaptive intelligence, multi-factor control, and crop-specific prediction logic.
Web-Based Sluice Gate System (Desnanjaya & Nugraha, 2021)	Combines **Arduino**, **ultrasonic sensor**, and **servo motor** for water gate control via web dashboard and SMS. Enables manual and automated modes for flood prevention. Cloud-based data logging and multi-user access.	Integrates **AI prediction**, **multi-sensor inputs (soil, water, forecast)**, and **fully autonomous actuation**. Closed-loop system—predicts irrigation need and controls gate automatically.	Web-controlled system is **semi-automated and reactive**, requiring human input. No **machine learning**, **rain integration**, or **adaptive irrigation scheduling**. Focused on monitoring, not prediction.

Autogation AWD-Based Smart Irrigation (Tolentino et al., 2021)	Implements **Alternate Wetting and Drying (AWD)**-based IoT irrigation using Arduino Mega, Zigbee, ultrasonic sensors, and stepper motor-controlled gates. Maintains water between −100 mm and +150 mm thresholds for paddy fields.	Enhances AWD logic with **AI-based predictive control**, **rain forecasting**, and **crop-stage adaptability** for proactive irrigation. Fully closes the loop (sensor → prediction → gate control).	The system is **rule-based and static**, lacks **adaptive learning**, **weather integration**, and **dynamic scheduling**. Cannot optimize water based on forecasted rainfall or field variability.
AWD Water Management & Hydrological Study (Aziz et al., 2021)	Examines AWD vs continuous flooding in 1200 m² paddy plots. AWD saved ~29.2% water, maintained stable soil moisture, and had no crop stress, but all measurements and irrigation were manual.	Automates AWD method using **AI-driven prediction**, **sensor feedback**, and **motorized gate actuation**. Integrates **rain forecast** and **crop-phase scheduling** for optimized water management.	Entirely **manual process** — no automation, sensors, or forecasting. Lacks real-time adaptation, predictive logic, or integration with smart irrigation infrastructure.
FAO-CROPWAT-8.0 Simulation for Rice Water Requirement (Singh & Verma, 2020)	Uses the **FAO CROPWAT-8.0** model with Penman–Monteith and crop coefficients to simulate evapotranspiration and irrigation needs for Dehradun. Provides monthly irrigation schedules based on historical climate data.	Replaces static simulation with **real-time ML-based irrigation prediction** using soil and weather sensors, plus **automated gate control** for live implementation.	Model is **offline and manual**, lacks **automation**, **sensor input**, and **forecast adaptability**. Focuses on theoretical planning, not dynamic irrigation execution.
IoT-based Smart Irrigation using DRL Optimization (Saikai et al., 2023)	Proposes a **Deep Reinforcement Learning (DRL)** model (REINFORCE algorithm) for optimizing irrigation scheduling within the APSIM crop simulator. Learns irrigation actions using simulated data, outperforming rule-based methods by up to 17%.	Implements **real-time DRL-style learning** in physical paddy environments via IoT sensors, gate motors, and weather APIs for adaptive, profit-oriented irrigation control.	Model is **simulation-only**, with **no real-world deployment, sensor integration, or hardware actuation**. Requires adaptation for field-level IoT control and dynamic weather integration.
Advanced Prediction of	Develops ML models (LSTM, SVM, ANN, RF,	Links **ML-based CWR prediction**	Web tool only — no automation or real-

Crop Water Requirements using ML Models (Nagappan et al., 2025)	GBM) to predict Evapotranspiration (ETo) and CWR using meteorological inputs (temp, humidity, solar radiation, rainfall, wind, sunshine). LSTM performed best (RMSE = 0.45 mm/day, $R^2 = 0.92$). Deployed via Flask web app for manual input and CWR output.	directly to **automated gate control**, using real-time sensors (soil, tank, weather) and forecast APIs for fully closed-loop irrigation.	time sensor integration. Relies on manual inputs, static Kc values, and offline execution; not field-deployable or paddy-specific.
Proper Predictions of the Water Fate in Agricultural Lands (Awad et al., 2021)	Compares empirical (CROPWAT) and hydrology-based (DRAINMOD) models for accurate estimation of Crop Water Requirements (CWR). Uses drainage simulations to adjust irrigation need based on water table, infiltration, and field drainage. DRAINMOD reduced CWR by ~59% compared to empirical models.	Converts **hydrological accuracy** into a **real-time ML-driven decision system**. Integrates live soil moisture, rainfall, and water-level sensors to predict irrigation need and actuate gates automatically.	**Simulation-based and manual**, not real-time or automated. Lacks **sensor integration**, **AI-based forecasting**, and **execution control**. Designed for wheat; not adaptable to paddy's standing-water conditions.

4. Theoretical Framework

The proposed research is grounded in three interconnected theoretical foundations: **precision agriculture**, **cyber-physical systems (CPS)**, and **machine learning for predictive modeling**. Together, these concepts provide a robust framework for designing an intelligent irrigation system for paddy cultivation.

Precision agriculture theory emphasizes the optimization of farming practices through site-specific data collection and analysis. In the context of paddy fields, this involves monitoring soil conditions, water levels, and climate data to ensure efficient irrigation. By applying precision agriculture principles, the project seeks to minimize water wastage while maintaining optimal soil moisture for crop growth.

The system also aligns with the **cyber-physical systems (CPS) paradigm**, where interconnected hardware and software components interact with the physical environment. Within this framework, the **perception layer** consists of sensors and APIs collecting soil, water, and weather data. The **processing layer** is the Raspberry Pi running machine learning models to generate irrigation decisions. The **actuation layer** is represented by the motorized water gate mechanism, while the **application layer** provides user-level insights and control. This layered CPS approach ensures adaptability, scalability, and real-time responsiveness.

Finally, **machine learning theories of regression and time-series forecasting** guide the predictive modeling process. By learning from historical soil and weather data, the model can estimate future water requirements, moving beyond reactive threshold-based systems. The integration of supervised learning and environmental factors positions the system as a proactive irrigation manager.

In summary, this research builds on theoretical underpinnings of precision agriculture, CPS, and machine learning to create a unified framework that enables real-time sensing, predictive decision-making, and automated water management for sustainable paddy cultivation.

5. Research Methodology

This study adopts a **design science approach**, focusing on the development and validation of a predictive model for irrigation management in paddy cultivation. The methodology is structured into four phases: system conceptualization, data collection, model development, and evaluation.

System Design

The proposed system is designed as a layered framework comprising three core components: sensing, processing, and actuation. The sensing layer will capture data from environmental factors such as soil moisture, water levels, rainfall detection, and weather forecasts obtained through free APIs. The processing layer will integrate these heterogeneous inputs and apply a machine learning–based predictive model to determine the water requirement at a given time. The actuation layer will then automatically regulate irrigation by controlling a motorized water gate, ensuring that water is released in accordance with the model's predictions. The system is conceptualized to operate in real time, supporting proactive irrigation decisions that reduce water wastage and maintain optimal soil conditions.

Data Collection

The data required for this study will include soil moisture levels, water availability, and weather parameters such as temperature and rainfall forecasts. Instead of relying solely on existing datasets, the researcher will identify the most relevant parameters by reviewing prior studies and consulting with an industry expert in agriculture. This collaboration will help tailor the dataset to the specific requirements of paddy cultivation in Sri Lanka. The dataset will then be constructed with the expert's input to ensure that it reflects real-world agricultural practices and conditions.

Model Development

The predictive model will be built using supervised learning techniques. Input features will include environmental parameters identified during the data collection phase. Multiple algorithms will be explored to determine which approach offers the best accuracy in estimating irrigation needs. Standard preprocessing methods such as normalization, feature selection, and data splitting will be applied. Model performance will be evaluated using appropriate statistical metrics, including **Root Mean Squared Error (RMSE)** and **R^2 score**, to determine predictive accuracy.

Evaluation

The evaluation stage will focus primarily on the **prediction accuracy** of the developed model. The results will be compared against baseline methods (such as threshold-based soil moisture prediction) to assess improvement. The emphasis will be on validating whether the model can consistently generate reliable predictions that align with expert expectations and real-world observations.

6. Expected Contributions

This research is expected to contribute both academically and practically to the field of smart irrigation and agricultural water management.

Academic Contributions

From an academic standpoint, the project advances the literature on precision agriculture by demonstrating how real-time environmental sensing and predictive modeling can be integrated into a unified framework for irrigation management. While many studies focus either on soil moisture prediction or automated irrigation, this work bridges the gap by combining data-driven machine learning with practical actuation. The research will also provide comparative insights into how different environmental parameters such as soil moisture, rainfall forecasts, and water availability affect predictive accuracy. These findings will add to the body of knowledge on feature selection and model design in agricultural machine learning applications.

Practical Contributions

In practical terms, the project offers a low-cost, scalable, and farmer-friendly solution tailored to paddy cultivation, which is vital for food security in Sri Lanka and many other regions. By enabling proactive water management, the system has the potential to reduce water wastage, improve crop yields, and increase resilience against climate variability. Its design, centered on affordability and accessibility, ensures that smallholder farmers can adopt it without the need for costly infrastructure. Furthermore, the collaboration with industry experts during dataset creation strengthens the practical relevance of the model, ensuring that its outputs are aligned with real-world agricultural practices.

In summary, the research contributes new academic insights into predictive irrigation modeling while delivering a practical technological solution with direct benefits for sustainable agriculture.

7. Timeline

September 2025

- **Week 1 (Sep 22 – Sep 28):** Introduction to the research module and topic confirmation. Begin reviewing existing studies to identify the research problem.
- **Week 2 (Sep 29 – Oct 5):** Conduct a detailed literature review and refine research problem identification.
- **Deliverable:** *Begin draft of Project Proposal.*

October 2025

- **Week 3 (Oct 6 – Oct 12):** Start proposal writing process.
- **Week 4 (Oct 13 – Oct 19):** Study Research Methods in AI/DS.
    - **Deliverable:** *Project Proposal Submission.*
- **Week 5 (Oct 20 – Oct 26):** Refine and validate research questions; finalize scope and objectives.
- **Week 6 (Oct 27 – Nov 2):** Engagement Week (no major submissions).
    - **Deliverable:** *Literature Review Draft Submission.*

November 2025

- **Week 7 (Nov 3 – Nov 9):** Define data requirements and identify primary/secondary data sources. Work on SPER and methodology documentation.
    - **Deliverables:** *SPER Form Submission* and *Research Methodology Submission.*
- **Week 8 (Nov 10 – Nov 16):** Address ethical and security issues. Prepare dataset plan with industry expert.
- **Week 9 (Nov 17 – Nov 23):** Evaluation Planning — define metrics and evaluation framework.
- **Week 10 (Nov 24 – Nov 30):** Proposal Poster Presentations (Part 1).
- **Week 11 (Dec 1 – Dec 7):** Proposal Poster Presentations (Part 2).

December 2025

- **Week 12 (Dec 8 – Dec 14):** Complete evaluation plan and finalize evaluation framework.
    - **Deliverable:** *Evaluation Plan Submission.*
- Begin procuring hardware components (soil moisture, tank level, rainfall sensors, motorized gate).
- Initial dataset preparation and hardware setup.

January 2026

- **Week 13 (Jan 19 – Jan 25):** Organize project artefacts (citations, concept mapping, project structure).
- Continue building dataset with industry expert; preprocess data (cleaning, normalization, feature extraction).
- Begin preliminary hardware assembly and integration tests.

February 2026

- **Week 14 (Jan 26 – Feb 1):** Continue thesis writing preparation.
- **Week 15 (Feb 2 – Feb 8):** Begin research paper writing drafts.
- **Week 16 (Feb 9 – Feb 15):** Develop predictive model; integrate hardware with software (functional prototype).
  - **Deliverable:** *Interim Progress Presentation with Functional Prototype (Feb 9).*
- **Week 17 (Feb 16 – Feb 22):** Refine prototype and continue system testing.

March 2026

- **Week 18 (Feb 23 – Mar 1):** Finalize predictive model with improved accuracy and performance.
- **Week 19 (Mar 2 – Mar 8):** Begin preparing final documentation.
- **Week 20 (Mar 9 – Mar 15):** Conduct internal viva practice sessions.
- **Week 21 (Mar 16 – Mar 22):** Participate in *Thesis Feedback Discussion.*
- **Week 22 (Mar 23 – Mar 29):** Apply feedback and finalize thesis and research paper drafts.

April 2026

- **Week 23 (Mar 30 – Apr 5):** Conduct final testing and evaluation of predictive model + full system validation.
- **Week 24 (Apr 6 – Apr 12):**
  - **Deliverables:** *IRP Thesis Submission* and *IRP Research Paper Submission.*
- Prepare for *Final Viva Presentation.*

8. References

1. https://www.researchgate.net/publication/235891972_Estimation_of_soil_moisture_in_paddy_field_using_Artificial_Neural_Networks
2. https://www.researchgate.net/publication/335358334_Generating_Soil_Salinity_Soil_Moisture_Soil_pH_from_Satellite_Imagery_and_Its_Analysis
3. https://www.sciencedirect.com/science/article/pii/S0957417422009563
4. https://iopscience.iop.org/article/10.1088/1755-1315/1109/1/012067

5. https://www.sciencedirect.com/science/article/pii/S0048969722021593
6. https://www.sciencedirect.com/science/article/pii/S2210537918302464
7. https://www.sciencedirect.com/science/article/pii/S0016706117321031
8. https://www.sciencedirect.com/science/article/pii/S095741742200269X
9. https://www.sciencedirect.com/science/article/pii/S0034425721001528
10. https://www.sciencedirect.com/science/article/pii/S0034425721004260
11. https://www.researchgate.net/publication/380288328_Automatic_gate_controlling_system_for_sustainable_agriculture_water_management_in_Sri_Lanka_A_review
12. https://www.researchgate.net/publication/349658847_Automated_Water-Gate_Controlling_System_for_Paddy_Fields
13. https://ieeexplore.ieee.org/document/9678410
14. https://ieeexplore.ieee.org/document/10841506
15. https://ieeexplore.ieee.org/document/9650409
16. https://www.researchgate.net/publication/354999767_Autogation_An_Alternate_Wetting_and_Drying-Based_Automatic_Irrigation_and_Paddy_Water_Level_Control_System_through_Internet_of_Things
17. https://www.researchgate.net/publication/381828405_Water_Management_and_Hydrological_Characteristics_of_Paddy-Rice_Fields_under_Alternate_Wetting_and_Drying_Irrigation_Practice_as_Climate_Smart_Practice_A_Review
18. https://www.researchgate.net/publication/340302993_Prediction_of_Rice_Water_Requirement_Using_FAO-CROPWAT-80_Model_in_Dehradun_Uttarakhand_India
19. https://arxiv.org/abs/2301.00899?utm_source=chatgpt.com
20. https://eprint.ijisrt.org/id/eprint/357/?utm_source=chatgpt.com
21. https://www.sciencedirect.com/science/article/pii/S2090447921000678